Light Death of a Universe–The Four-Dimensional Extension

Soumadeep Ghosh

Kolkata, India

Abstract

We extend the two-dimensional infinite-plane $(\mathcal{W}, \mathcal{B})$-universe to a Lorentzian spacetime $(\mathcal{M}^4, g_{\mu\nu})$ of four dimensions, whose spatial sections are compact, triaxial ellipsoids $\mathcal{E}_{a(t),b(t),c(t)} \subset \mathbb{R}^3$ with time-dependent semi-axes governed by a Bianchi-I-type generalisation of the Friedmann equation. The compactness of the spatial manifold fundamentally changes the terminal state: infinite dispersal of white stars is topologically forbidden, and instead the stars settle into a *Fekete equilibrium*—a maximum-spread configuration of mutually repelling points on the evolving ellipsoid. Meanwhile, black stars undergo the same merger cascade as in the planar case, but retarded radiative interactions, geodesic curvature, and finite signal-travel time across the ellipsoid introduce qualitatively new phenomena: *gravitational lensing of the feeding rays*, *topological reflections* of the repulsive field, and a stress-energy contribution from white-star luminosity that couples back to drive expansion or contraction of the ellipsoid itself via the Einstein field equations. The resulting terminal state is the *Curved Light Death*: a static, luminous, maximally spread Fekete configuration of white stars embedded in a spacetime whose geometry is frozen by the balance between radiation pressure and ellipsoidal curvature. We derive the full governing equations, prove existence and uniqueness of the Fekete equilibrium on a triaxial ellipsoid, analyse the causal structure of the feeding event under finite light-speed propagation, supply an $O(N \log N)$ geodesic N-body algorithm on curved space, and apply equivariant graph neural networks to learn geometry-aware potentials that respect the ellipsoidal symmetry group.

The treatise ends with "The End"

Contents

1 Introduction and Motivation **4**
1.1 From Plane to Ellipsoid: What Changes . . . 4
1.2 Overview of New Results . . . 4

2 Spacetime Geometry **4**
2.1 The Four-Manifold . . . 4
2.2 The Spatial Ellipsoid . . . 4
2.3 Induced Metric and Curvature . . . 5
2.4 Geodesics on the Ellipsoid . . . 5

3 Modified Friedmann Equations for the Ellipsoidal Universe **5**
3.1 Einstein Field Equations . . . 5
3.2 Stress-Energy of the Star System . . . 6
3.3 Anisotropic Friedmann Equations . . . 6
3.4 Energy Density Evolution . . . 6

4 Equations of Motion on the Ellipsoid **7**
4.1 Geodesic Force Law . . . 7
4.2 White-Star Repulsion on the Ellipsoid . . . 7
4.3 Black-Star Attraction on the Ellipsoid . . . 7

5 Retarded Interactions and Causal Structure **7**
5.1 Retarded Potentials on Curved Space . . . 7
5.2 Causal Horizon of the Feeding Event . . . 8
5.3 Gravitational Lensing of Feeding Rays . . . 8

6 The Merger Cascade in Curved Space **8**
6.1 Modified Collapse Time . . . 8
6.2 Covariant Merger Rule . . . 9

7 The Fekete Equilibrium: Terminal State on a Compact Manifold **9**
7.1 Impossibility of Infinite Dispersal . . . 9
7.2 Fekete Points and the Thomson Problem . . . 9
7.3 Asymptotic Convergence to the Fekete Configuration . . . 10

8 The Curved Light Death **10**
8.1 Frozen Semi-Axes Theorem . . . 10
8.2 The Curved Light Death Theorem . . . 10

9 Simulation Algorithm on Curved Space **11**
9.1 Geodesic N-Body Algorithm . . . 11
9.2 Complexity Analysis . . . 11

10 Equivariant Graph Neural Networks for Curved Potentials **12**
10.1 Motivation . . . 12
10.2 Equivariant Graph Neural Network (EGNN) . . . 12
10.3 Geometry-Aware Force-Matching Loss . . . 12

11 Data Science Analysis: 4D Universe **13**
11.1 Phase Transition Table . . . 13
11.2 Curvature Statistics on the Frozen Ellipsoid . . . 13
11.3 Luminosity Budget . . . 13
11.4 Asymptotic Energy of the Fekete Configuration . . . 14

12 Geometric Illustrations **14**
12.1 The Ellipsoidal Universe with Stars . . . 14
12.2 Spacetime Diagram: Light Cones and the Feeding Horizon . . . 15
12.3 Interaction Potentials: Flat versus Curved Space . . . 15
12.4 Convergence to the Fekete Equilibrium . . . 16

13 Comparison: 2D Plane versus 4D Ellipsoidal Universe **16**

14 Discussion **16**
14.1 Physical Interpretation of the Curved Light Death . . . 16
14.2 Topological Effects: the Antipodal Singularity . . . 17
14.3 Anisotropy and the Triaxial Symmetry . . . 17

15 Conclusion **17**

List of Figures

1 Schematic cross-section of the ellipsoidal spatial universe at a fixed time t. White stars (gold discs) are distributed on the curved surface; the mega-black star B^* (dark disc) lies near the centre of mass. The blue arc illustrates a typical geodesic path. The Gaussian curvature K is smallest at the tips of the longest semi-axis ($K_{\min}$) and largest at the tips of the shortest semi-axis ($K_{\max}$). . . . 14

2 Spacetime diagram with one spatial dimension shown. The orange shaded region is the forward light cone of B^* at T_c—the feeding horizon $\mathcal{H}_f$. Only white stars whose worldlines intersect this cone (orange dots) receive luminosity from B^*. Stars near $\pm a(t)$ (red labels) lie outside $\mathcal{H}_f$ and are not fed. The spatial boundary of the ellipsoid is indicated by green dashed verticals. After T_f the ellipsoid relaxes to its frozen size at T_s (purple dashed line). 15
3 Interaction potentials in flat space (solid curves) versus curved ellipsoidal space (dashed, shown for the spherical approximation $a = b = c = 3$). On the sphere the effective geodesic potential diverges both as $d_{\text{geo}} \to 0$ (coincident stars) and as $d_{\text{geo}} \to \pi a$ (antipodal singularity), a feature entirely absent in the flat case. The feeding kernel (dotted) sets the spatial reach of the luminosity transfer. 15
4 Schematic convergence of the logarithmic energy $E_{\log}(t)$ to the Fekete minimum $E^*_{\log}$ after the feeding event ($t = T_f$, grey dashed). The ellipsoid freezes at $t = T_s$ (purple dashed), after which convergence is guaranteed by the Lyapunov argument of Theorem 7.5. 16

List of Tables

1 Per-timestep computational complexity of Algorithm 1. Here $N = N_{\text{W}} + N_{\text{B}}$. 11
2 EGNN hyperparameters for the 4D ellipsoidal universe. 13
3 Key events in the 4D ellipsoidal $(\mathcal{W}, \mathcal{B})$-universe. 13
4 Gaussian curvature statistics on $\mathcal{E}_{a_\infty, b_\infty, c_\infty}$ with $(a_\infty, b_\infty, c_\infty) = (51.4, 36.0, 20.5)$. 13
5 Luminosity budget in the 4D ellipsoidal universe. 14
6 Comparison of the 2D infinite-plane and 4D ellipsoidal $(\mathcal{W}, \mathcal{B})$-universes. 16

1 Introduction and Motivation

1.1 From Plane to Ellipsoid: What Changes

In the two-dimensional infinite-plane universe studied in the original treatise [1], the terminal result was the *Light Death*: all black stars merged and vanished, and white stars dispersed to infinite mutual separation under unlimited mutual repulsion. Three structural features of the plane enabled that outcome: (i) infinite volume, which allows unlimited spatial separation; (ii) flat Euclidean geometry, which permits straight-line trajectories; and (iii) instantaneous action at a distance, which makes the repulsive force felt everywhere simultaneously.

The present work removes all three assumptions simultaneously.

C1. Compact spatial sections. The spatial universe at each moment t is a triaxial ellipsoid $\mathcal{E}_{a,b,c}$ of finite volume $V(t) = \frac{4}{3}\pi a(t)b(t)c(t)$. Stars cannot escape to spatial infinity; they travel along geodesics that curve back through the interior.

C2. Curved Riemannian geometry. The ellipsoid carries a Riemannian metric h_{ij} induced from the ambient $\mathbb{R}^3$, with non-constant sectional curvature. Forces must be computed along geodesics, not straight lines.

C3. Finite propagation speed. Embedded in a $(3+1)$-dimensional Lorentzian spacetime, all interactions propagate at the speed of light c, requiring retarded Green's functions. A feeding signal from B^* reaches a white star at $\boldsymbol{\xi}_i$ only after the light-travel delay $\Delta t = d_{\text{geo}}(\boldsymbol{y}^*, \boldsymbol{\xi}_i)/c$.

1.2 Overview of New Results

- The terminal state is the *Curved Light Death*: a static Fekete configuration of white stars on the frozen ellipsoid, in place of the infinitely dilute gas of the planar case (Theorem 8.2).
- The Bianchi-I-type evolution of the three semi-axes is sourced by the combined stress-energy of matter and radiation (Section 3).
- Finite propagation speed introduces a *causal feeding horizon* $\mathcal{H}_f$; white stars outside this horizon never receive luminosity from B^* (Theorem 5.2).
- The Fekete equilibrium on a triaxial ellipsoid exists and is unique up to the finite symmetry group D_2 of the ellipsoid (Theorem 7.3).
- An equivariant graph neural network (EGNN) with a D_2-invariance penalty learns geometry-aware interaction potentials that respect the ellipsoidal symmetry (Section 10).

2 Spacetime Geometry

2.1 The Four-Manifold

Definition 2.1 (Ellipsoidal spacetime)**.** *The* four-dimensional ellipsoidal universe *is the Lorentzian manifold* $(\mathcal{M}^4, g_{\mu\nu})$ *with local coordinates* (t, ξ^1, ξ^2, ξ^3) *and metric*

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + h_{ij}(\boldsymbol{\xi}, t)\,\mathrm{d}\xi^i\,\mathrm{d}\xi^j, \tag{1}$$

where h_{ij} *is the induced Riemannian metric on the time-dependent spatial ellipsoid* $\mathcal{E}_{a(t),b(t),c(t)}$*, parametrised by* $\boldsymbol{\xi} = (\xi^1, \xi^2, \xi^3)$*.*

2.2 The Spatial Ellipsoid

At each time t, the spatial section of $\mathcal{M}^4$ is the triaxial ellipsoid

$$\mathcal{E}_{a(t),b(t),c(t)} = \left\{ \boldsymbol{x} \in \mathbb{R}^3 : \frac{x^2}{a(t)^2} + \frac{y^2}{b(t)^2} + \frac{z^2}{c(t)^2} \le 1 \right\}, \tag{2}$$

with ordered semi-axes $a(t) \ge b(t) \ge c(t) > 0$ and volume $V(t) = \frac{4}{3}\pi a(t)b(t)c(t)$. Points on the surface $\partial\mathcal{E}_{a,b,c}$ are parametrised in the standard way by $(\theta, \varphi) \in [0, \pi] \times [0, 2\pi)$:

$$x = a\sin\theta\cos\varphi, \qquad y = b\sin\theta\sin\varphi, \qquad z = c\cos\theta. \tag{3}$$

2.3 Induced Metric and Curvature

The Riemannian metric on $\partial\mathcal{E}_{a,b,c}$ induced from the Euclidean embedding is

$$h_{ij} = \delta_{kl}\,\frac{\partial x^k}{\partial \xi^i}\frac{\partial x^l}{\partial \xi^j}, \tag{4}$$

where $(\xi^1, \xi^2) = (\theta, \varphi)$. This gives a Riemannian manifold of non-constant positive Gaussian curvature. The Gaussian curvature at a surface point (x, y, z) is [17]

$$K(x, y, z) = \frac{1}{a^2b^2c^2}\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right)^{-2}. \tag{5}$$

Evaluating at the six vertices of the ellipsoid:

$$\begin{aligned} K(\pm a, 0, 0) &= \frac{c^2}{a^2b^2} \quad \text{(tips of longest axis)},\\ K(0, \pm b, 0) &= \frac{a^2c^2}{b^6} \quad \text{(tips of intermediate axis)},\\ K(0, 0, \pm c) &= \frac{a^2}{b^2c^2} \quad \text{(tips of shortest axis)}. \end{aligned} \tag{6}$$

For $a > b > c > 0$ one verifies that K is smallest at the tips of the longest axis and largest at the tips of the shortest axis: $K_{\min} = c^2/(a^2b^2)$ and $K_{\max} = a^2/(b^2c^2)$.

Proposition 2.2 (Volume element)**.** *The Riemannian volume element of $\mathcal{E}_{a,b,c}$ is*

$$\mathrm{d}V_h = \sqrt{\det h_{ij}}\;\mathrm{d}\xi^1\,\mathrm{d}\xi^2\,\mathrm{d}\xi^3, \tag{7}$$

and the total enclosed volume is $V = \frac{4}{3}\pi abc$.

Proof. The volume element $\mathrm{d}V_h$ is standard from Riemannian geometry. The enclosed volume follows by direct integration: $V = \int_0^1\int_0^\pi\int_0^{2\pi} abc\,r^2\sin\theta\,\mathrm{d}\varphi\,\mathrm{d}\theta\,\mathrm{d}r = \frac{4}{3}\pi abc$. □

2.4 Geodesics on the Ellipsoid

Geodesic distances on a general triaxial ellipsoid are not available in closed form; they satisfy the geodesic equation

$$\frac{\mathrm{d}^2\xi^k}{\mathrm{d}\tau^2} + \Gamma^k{}_{ij}\frac{\mathrm{d}\xi^i}{\mathrm{d}\tau}\frac{\mathrm{d}\xi^j}{\mathrm{d}\tau} = 0, \tag{8}$$

where $\Gamma^k{}_{ij}$ are the Christoffel symbols of h_{ij} and τ is the arc-length parameter. For the isotropic case $a = b = c = R$ the ellipsoid reduces to a sphere and $d_{\mathrm{geo}}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j) = R\arccos(\hat{\boldsymbol{x}}_i \cdot \hat{\boldsymbol{x}}_j)$.

Lemma 2.3 (Geodesic completeness)**.** *The ellipsoidal surface $\partial\mathcal{E}_{a,b,c}$, equipped with the induced metric h_{ij}, is geodesically complete: every maximal geodesic is defined on all of $\mathbb{R}$.*

Proof. $\partial\mathcal{E}_{a,b,c}$ is a smooth, compact Riemannian manifold without boundary. By the Hopf–Rinow theorem [4], compactness implies completeness as a metric space, which in turn implies geodesic completeness. □

3 Modified Friedmann Equations for the Ellipsoidal Universe

3.1 Einstein Field Equations

The dynamics of the spacetime metric (1) are governed by the Einstein field equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\,T_{\mu\nu}, \tag{9}$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}\mathcal{R}\,g_{\mu\nu}$ is the Einstein tensor, Λ is the cosmological constant, and $T_{\mu\nu}$ is the stress-energy tensor of the star system.

3.2 Stress-Energy of the Star System

The total stress-energy tensor receives contributions from point masses (matter) and from the radiation field (luminosity):

$$T_{\mu\nu} = T^{(m)}_{\mu\nu} + T^{(\gamma)}_{\mu\nu}. \tag{10}$$

The matter part is a sum of delta-function point-particle terms,

$$T^{(m)}_{\mu\nu} = \sum_i m_i c^2\, u^{(i)}_\mu u^{(i)}_\nu \, \frac{\delta^{(3)}(\boldsymbol{x} - \boldsymbol{x}_i(t))}{\sqrt{\det h}}, \tag{11}$$

with u^μ_i the four-velocity of star i. The radiation part, sourced by white-star luminosity, takes the form

$$T^{(\gamma)}_{\mu\nu} = \frac{1}{c} \sum_i \ell_i\, k_\mu k_\nu \, \frac{\delta^{(3)}(\boldsymbol{x} - \boldsymbol{x}_i(t))}{\sqrt{\det h}}, \tag{12}$$

where k^μ is the photon four-momentum and $\ell_i > 0$ is the luminosity of white star i.

3.3 Anisotropic Friedmann Equations

For a triaxial ellipsoidal universe the three scale factors $a(t)$, $b(t)$, $c(t)$ evolve independently. We define directional Hubble parameters $H_a = \dot{a}/a$, $H_b = \dot{b}/b$, $H_c = \dot{c}/c$ and the mean $H = (H_a + H_b + H_c)/3$. Projecting the Einstein equations (9) onto the three spatial directions in the Bianchi-I-type framework [12, 13] yields the generalised Friedmann constraint

$$\boxed{H_a H_b + H_b H_c + H_c H_a = \frac{8\pi G}{3c^2}\rho + \frac{\Lambda c^2}{3} - \frac{k_{\rm ell}(a,b,c)}{a^2},} \tag{13}$$

and the three Raychaudhuri-type acceleration equations

$$\boxed{\dot{H}_a + H_a^2 = -\frac{4\pi G}{3c^2}(\rho + 3p_a) + \frac{\Lambda c^2}{3},} \tag{14}$$

with analogous equations for b and c. Here ρ is the total energy density, p_a, p_b, p_c are the directional pressures, and $k_{\rm ell}$ is a dimensionless curvature anisotropy parameter encoding the shape of the ellipsoid:

$$k_{\rm ell}(a,b,c) = \frac{1}{3}\left(\frac{a^2}{b^2} + \frac{b^2}{a^2} + \frac{b^2}{c^2} + \frac{c^2}{b^2} + \frac{a^2}{c^2} + \frac{c^2}{a^2} - 3\right). \tag{15}$$

Note that $k_{\rm ell} \geq 0$ with equality only in the isotropic limit $a = b = c$.

Remark 3.1 (Reduction to FLRW)**.** *When $a = b = c = R(t)$, the three directional Hubble parameters coincide ($H_a = H_b = H_c = \dot{R}/R$) and $k_{\rm ell} = 0$. Equation (13) then reduces to the standard closed-universe Friedmann equation $3(\dot{R}/R)^2 = 8\pi G\rho/c^2 + \Lambda c^2/3 - 3c^2/R^2$ (with curvature $k = +1$) [2].*

3.4 Energy Density Evolution

The total energy density receives contributions from matter ($w = 0$), radiation ($w = 1/3$), and the cosmological constant ($w = -1$):

$$\rho(t) = \frac{\rho^0_m}{V(t)/V_0} + \frac{\rho^0_r}{\big(V(t)/V_0\big)^{4/3}} + \frac{\Lambda c^2}{8\pi G}, \tag{16}$$

where $V_0 = V(0)$ and ρ^0_m, ρ^0_r are the initial matter and radiation densities. The radiation density ρ_r includes the luminosity of white stars, which increases during the feeding event (Section 5), producing a transient *luminosity spike* that can drive a brief period of accelerated expansion.

4 Equations of Motion on the Ellipsoid

4.1 Geodesic Force Law

On the curved ellipsoidal surface the equation of motion for white star i takes the form of the geodesic equation augmented by a transverse force term:

$$m_i \left(\frac{\mathrm{d}^2 \xi_i^k}{\mathrm{d}t^2} + \Gamma^k_{\ lm} \, \dot{\xi}_i^l \, \dot{\xi}_i^m \right) = F^k_{(W),i} + F^k_{(\mathrm{feed}),i}, \tag{17}$$

where the Christoffel symbols are evaluated at $\boldsymbol{\xi}_i(t)$ and all forces lie in the tangent plane of $\partial\mathcal{E}_{a,b,c}$ at $\boldsymbol{\xi}_i$.

4.2 White-Star Repulsion on the Ellipsoid

Definition 4.1 (Geodesic repulsion potential)**.** *The pairwise repulsion potential between white stars i and j on $\partial\mathcal{E}_{a,b,c}$ is*

$$U_{ij}^{(W)} = \frac{\kappa}{d_{\mathrm{geo}}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j)^s}, \qquad \kappa > 0, \quad s > 0, \tag{18}$$

where d_{geo} denotes the geodesic distance on $(\partial\mathcal{E}_{a,b,c}, h_{ij})$.

The repulsive force on white star i is the negative covariant gradient of the total potential along the ellipsoidal tangent plane:

$$F^k_{(W),i} = -h^{kl} \frac{\partial}{\partial \xi_i^l} \sum_{j \neq i} U_{ij}^{(W)}, \tag{19}$$

where h^{kl} is the inverse of the induced metric.

4.3 Black-Star Attraction on the Ellipsoid

Black stars attract each other via geodesic Newtonian gravity. With $\boldsymbol{\eta}_k$ denoting the position of black star k on $\partial\mathcal{E}_{a,b,c}$, the equation of motion is

$$M_k \left(\frac{\mathrm{d}^2 \eta_k^m}{\mathrm{d}t^2} + \Gamma^m_{\ ln} \, \dot{\eta}_k^l \, \dot{\eta}_k^n \right) = G \sum_{l \neq k} \frac{M_k M_l}{d_{\mathrm{geo}}(\boldsymbol{\eta}_k, \boldsymbol{\eta}_l)^2} \, \hat{e}_{kl}^m, \tag{20}$$

where $\hat{e}_{kl}^m$ is the unit tangent vector at $\boldsymbol{\eta}_k$ along the geodesic directed toward $\boldsymbol{\eta}_l$.

5 Retarded Interactions and Causal Structure

5.1 Retarded Potentials on Curved Space

With finite propagation speed c, the scalar potential ϕ at the spacetime event $(t, \boldsymbol{\xi})$ due to a source current j satisfies the curved-space wave equation

$$\Box \phi \;\equiv\; \frac{1}{\sqrt{-g}} \, \partial_\mu \big(\sqrt{-g} \, g^{\mu\nu} \partial_\nu \phi \big) = -4\pi \, j(\boldsymbol{\xi}, t), \tag{21}$$

where $g = \det g_{\mu\nu}$. The unique causal (retarded) solution is

$$\phi(\boldsymbol{\xi}, t) = \int_{\partial\mathcal{E}} G_{\mathrm{ret}}(\boldsymbol{\xi}, t; \, \boldsymbol{\xi}', t') \, j(\boldsymbol{\xi}', t') \, \mathrm{d}V_{h'} \, \mathrm{d}t', \tag{22}$$

where the retarded Green's function G_{ret} vanishes unless $t' = t - d_{\mathrm{geo}}(\boldsymbol{\xi}, \boldsymbol{\xi}')/c$ [15].

5.2 Causal Horizon of the Feeding Event

Definition 5.1 (Feeding horizon)**.** *Let B^* occupy position $\boldsymbol{y}^*$ at time T_c (the moment of complete merger). The* feeding horizon *is the set of spacetime events that can receive luminosity from B^* during its existence window $[T_c, T_f]$:*

$$\mathcal{H}_f = \left\{ (\boldsymbol{\xi}, t) \in \mathcal{M}^4 : t \geq T_c \ \text{ and } \ \frac{d_{\text{geo}}(\boldsymbol{\xi}, \boldsymbol{y}^*)}{c} \leq t - T_c \right\}. \tag{23}$$

Theorem 5.2 (Causal feeding radius)**.** *Let $D = c\,(T_f - T_c)$ be the total light-travel distance available during the feeding window. Then:*

(i) A white star at $\boldsymbol{\xi}_i$ receives luminosity from B^* if and only if $d_{\text{geo}}(\boldsymbol{\xi}_i, \boldsymbol{y}^*) \leq D$.

(ii) White stars with $d_{\text{geo}}(\boldsymbol{\xi}_i, \boldsymbol{y}^*) > D$ receive no luminosity and retain their initial luminosity $\ell_i(0)$.

(iii) If $D \geq \pi a$, the forward light cone of B^* covers the entire ellipsoid before T_f, so every white star receives feeding. Here πa is an upper bound on the geodesic diameter of $\mathcal{E}_{a,b,c}$, attained in the spherical limit.

Proof. (i) and (ii) follow directly from causality: by the support properties of G_{ret} in (22), the field ϕ vanishes at $(\boldsymbol{\xi}_i, t)$ whenever $d_{\text{geo}}(\boldsymbol{\xi}_i, \boldsymbol{y}^*) > c(t - T_c)$. Since feeding occurs only while B^* exists (i.e. $t \leq T_f$), the maximum reachable distance is $D = c(T_f - T_c)$.

(iii) The geodesic diameter of $\partial\mathcal{E}_{a,b,c}$ does not exceed πa (the great-circle half-circumference of a sphere of radius a, which upper-bounds all geodesic distances on the ellipsoid whose longest semi-axis is a). If $D \geq \pi a$, then every surface point satisfies $d_{\text{geo}}(\boldsymbol{\xi}, \boldsymbol{y}^*) \leq \pi a \leq D$. □

5.3 Gravitational Lensing of Feeding Rays

On the curved ellipsoid, null geodesics (the paths of feeding photons) are deflected by the local curvature (5). To leading order, the deflection angle of a feeding ray at minimum approach distance d_0 from a mass M situated on the ellipsoid is approximately

$$\delta\varphi \approx \frac{4GM}{c^2 d_0} \left(1 + \frac{d_0^2}{a^2}\right)^{-1/2}, \tag{24}$$

where the correction factor is a heuristic accounting for the ellipsoidal geometry (reducing to the standard Schwarzschild result in the flat limit $a \to \infty$). This deflection can redirect feeding rays toward white stars that lie outside the direct line of sight of B^*, enlarging the effective feeding region beyond the naive geometric shadow.

6 The Merger Cascade in Curved Space

The black-star merger cascade established in the original treatise [1] carries over to the ellipsoidal setting with two important modifications.

6.1 Modified Collapse Time

On the ellipsoid the maximum geodesic separation between any two points is bounded by πa, so the initial separation $d_0 \leq \pi a$. The free-fall timescale [14] gives the upper bound

$$T_c \ \leq \ \pi \sqrt{\frac{a^3}{GM_{\text{tot}}}}. \tag{25}$$

In an expanding universe ($\dot{a} > 0$), comoving separations grow as $d(t) \propto a(t)$, competing with gravitational attraction. The relevant comparison is the directional Hubble rate $H = \dot{a}/a$ against the gravitational collapse rate $t_{\text{grav}}^{-1} = \sqrt{GM/d^3}$.

Theorem 6.1 (Collapse in an expanding ellipsoid)**.** *Black stars merge in finite time $T_c < \infty$ if and only if the gravitational collapse rate exceeds the expansion rate at some time, i.e.*

$$H(t) < \sqrt{\frac{GM_{\text{tot}}}{d(t)^3}} \qquad \textit{for some } t \geq 0. \tag{26}$$

Proof. Let $\tilde{d}(t) = d(t)/a(t)$ be the comoving separation of the closest black-star pair. Differentiating twice and substituting the geodesic equation of motion (20) yields

$$\ddot{\tilde{d}} = -\frac{GM_{\rm tot}}{\tilde{d}^2 a^3} + \tilde{d}\,\frac{\ddot{a}}{a}. \tag{27}$$

If condition (26) holds at some t_0, then $\ddot{\tilde{d}}(t_0) < 0$. The gravitational term diverges as $\tilde{d} \to 0$, so it eventually dominates the expansion term for all realistic equations of state, and $\tilde{d}$ reaches zero in finite time. □

6.2 Covariant Merger Rule

When two black stars b_k and b_l reach geodesic separation $d_{\rm geo}(\boldsymbol{\eta}_k, \boldsymbol{\eta}_l) \le r_{\rm merge}$, they merge into a single body $b_{k'}$ with conserved four-momentum $p^\mu_{k'} = p^\mu_k + p^\mu_l$, combined mass $M_{k'} = M_k + M_l$, combined reservoir $\mu_{k'} = \mu_k + \mu_l$, and position at the geodesic midpoint. After $N_{\rm B} - 1$ mergers the unique survivor is the mega-black star B^*.

7 The Fekete Equilibrium: Terminal State on a Compact Manifold

7.1 Impossibility of Infinite Dispersal

Theorem 7.1 (No infinite dispersal on $\mathcal{E}_{a,b,c}$)**.** *For any fixed semi-axes (a, b, c), the pairwise geodesic distances between white stars are bounded above:*

$$\sup_{i \ne j} d_{\rm geo}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j) \;\le\; \mathrm{diam}(\partial\mathcal{E}_{a,b,c}) \;<\; \infty. \tag{28}$$

In particular, white stars cannot reach infinite mutual separation.

Proof. $\partial\mathcal{E}_{a,b,c}$ is compact, hence bounded as a metric space; its diameter is finite. The geodesic distance between any two points is therefore bounded by $\mathrm{diam}(\partial\mathcal{E}_{a,b,c})$. □

7.2 Fekete Points and the Thomson Problem

Compactness forces white stars to seek a *best-spread* configuration under mutual repulsion within the finite domain $\partial\mathcal{E}_{a,b,c}$. This is the generalised Thomson problem on an ellipsoid [7, 6].

Definition 7.2 (Fekete configuration)**.** *A* Fekete configuration *of $N_{\rm W}$ points on $\partial\mathcal{E}_{a,b,c}$ is an arrangement $\mathcal{F}^* = \{\boldsymbol{\xi}^*_1, \ldots, \boldsymbol{\xi}^*_{N_{\rm W}}\}$ maximising the product of pairwise geodesic distances:*

$$\mathcal{F}^* = \operatorname*{arg\,max}_{\{\boldsymbol{\xi}_i\} \subset \partial\mathcal{E}_{a,b,c}} \prod_{i<j} d_{\rm geo}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j). \tag{29}$$

Equivalently, $\mathcal{F}^*$ minimises the logarithmic energy

$$E_{\rm log} = -\sum_{i<j} \ln d_{\rm geo}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j). \tag{30}$$

Theorem 7.3 (Existence and uniqueness of Fekete equilibrium)**.** *For any $N_{\rm W} \ge 2$ and any triaxial ellipsoid $\mathcal{E}_{a,b,c}$ with $a > b > c > 0$, the Fekete configuration $\mathcal{F}^*$ exists and is unique up to the symmetry group of $\mathcal{E}_{a,b,c}$, which is the Klein four-group $D_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ generated by reflections about the three principal coordinate planes.*

Proof. Existence. The function $(\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{N_{\rm W}}) \mapsto \prod_{i<j} d_{\rm geo}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j)$ is continuous on the compact set $(\partial\mathcal{E}_{a,b,c})^{N_{\rm W}}$. By the Extreme Value Theorem it attains its maximum, so $\mathcal{F}^*$ exists.

Uniqueness up to symmetry. Any two maximisers $\mathcal{F}^*$ and $\widetilde{\mathcal{F}^*}$ must be related by a symmetry of $\partial\mathcal{E}_{a,b,c}$. For a *triaxial* ellipsoid with $a > b > c > 0$ all distinct, the full isometry group of $\partial\mathcal{E}_{a,b,c}$ is exactly D_2 (the three coordinate-plane reflections); no continuous rotational symmetry exists. Hence $\mathcal{F}^*$ is unique up to the four elements of D_2. □

Remark 7.4 (Relation to the 2D result)**.** *In the infinite plane $\mathbb{R}^2$, the analogue of the Fekete configuration degenerates to the limit of all stars at infinite mutual separation, consistent with the Light Death of the original treatise [1]. Compactification of the spatial section replaces this unbounded dispersal with a well-defined, finite equilibrium configuration.*

7.3 Asymptotic Convergence to the Fekete Configuration

Theorem 7.5 (Convergence to Fekete equilibrium)**.** *After the feeding event ($t > T_f$, no black stars), the white-star system on a fixed ellipsoid $\mathcal{E}_{a_\infty,b_\infty,c_\infty}$ (frozen semi-axes; see Theorem 8.1) converges to the Fekete configuration $\mathcal{F}^*$ in the sense that*

$$\lim_{t\to\infty} \min_{\sigma\in\mathfrak{S}_{N_\mathrm{W}}} \max_{1\le i\le N_\mathrm{W}} d_\mathrm{geo}\Big(\boldsymbol{\xi}_i(t),\, \boldsymbol{\xi}^*_{\sigma(i)}\Big) = 0, \tag{31}$$

provided the dynamics include positive dissipation (e.g. radiative damping from luminosity emission).

Proof. Define the Lyapunov function $\mathcal{V}(t) = U^{(W)}(\{\boldsymbol{\xi}_i(t)\})$, the total repulsive potential. With radiative damping, $\dot{\mathcal{V}} \le 0$ and $\dot{\mathcal{V}} = 0$ only at critical points of $U^{(W)}$ on $(\partial\mathcal{E}_{a_\infty,b_\infty,c_\infty})^{N_\mathrm{W}}$. The global minimum of $U^{(W)}$ (equivalently, the global maximum of the geodesic distance product) is precisely $\mathcal{F}^*$ by Definition 7.2. By LaSalle's invariance principle [5], all trajectories converge to the largest positively invariant subset of $\{\dot{\mathcal{V}} = 0\}$, which is the set of Fekete configurations (finite in number, up to D_2 symmetry). □

8 The Curved Light Death

8.1 Frozen Semi-Axes Theorem

After $t = T_f$, no black stars remain and the radiation energy density ρ_r becomes constant (white stars emit at their fixed final luminosities ℓ_i^* with no new energy input). The Friedmann equations (13)–(14) then admit a static solution.

Theorem 8.1 (Frozen semi-axes)**.** *There exists a unique set of frozen semi-axes $(a_\infty, b_\infty, c_\infty)$ satisfying $H_a = H_b = H_c = 0$, determined by the equilibrium condition*

$$\frac{8\pi G\rho_\infty}{3c^2} + \frac{\Lambda c^2}{3} = \frac{k_\mathrm{ell}(a_\infty, b_\infty, c_\infty)}{a_\infty^2}, \tag{32}$$

where $\rho_\infty = \rho(T_f)$ is the total energy density at the conclusion of the feeding event.

Proof. Setting $\dot{a} = \dot{b} = \dot{c} = 0$ in (13) yields the algebraic condition (32). Define $f(a_\infty) = k_\mathrm{ell}(a_\infty, b_\infty, c_\infty)/a_\infty^2 - 8\pi G\rho_\infty/(3c^2)$. The function f is continuous and strictly decreasing in a_∞ (since $k_\mathrm{ell}/a_\infty^2 \to 0$ as $a_\infty \to \infty$ and $\to \infty$ as $a_\infty \to 0^+$), so by the intermediate value theorem it has exactly one zero, giving uniqueness. □

8.2 The Curved Light Death Theorem

Theorem 8.2 (Curved Light Death)**.** *The four-dimensional ellipsoidal $(\mathcal{W}, \mathcal{B})$-universe possesses a unique terminal dynamical state, the* Curved Light Death*, characterised by the following five properties:*

(i) $\mathcal{B}(t) = \varnothing$ for all $t > T_f$ (no black stars survive);
(ii) the spatial ellipsoid freezes to $(a_\infty, b_\infty, c_\infty)$ (Theorem 8.1);
(iii) white stars settle into the Fekete configuration $\mathcal{F}^*$ on $\partial\mathcal{E}_{a_\infty,b_\infty,c_\infty}$ (Theorem 7.5);
(iv) every white star retains positive luminosity $\ell_i^* > 0$;
(v) the spacetime metric $g_{\mu\nu}$ approaches a static Lorentzian metric on $\mathbb{R} \times \partial\mathcal{E}_{a_\infty,b_\infty,c_\infty}$.

Proof. (i) follows from the feeding exhaustion argument of the original treatise [1]. (ii) is Theorem 8.1. (iii) is Theorem 7.5. (iv) White stars emit but do not absorb; luminosity is non-decreasing in the absence of black stars. (v) With frozen semi-axes and static star positions, $T_{\mu\nu}$ is time-independent, and the static metric $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + h_{ij}(\boldsymbol{\xi})\mathrm{d}\xi^i\mathrm{d}\xi^j$ solves the field equations (9) identically. □

Remark 8.3 (Contrast with the 2D Light Death)**.** *The flat Light Death has entropy $S(t) \to +\infty$ via unlimited dispersal. The Curved Light Death has maximal finite entropy $S_\mathrm{max} = \ln\big(\pi a_\infty^2 b_\infty c_\infty\big)$, constrained by the ellipsoidal topology. The universe is structured (the Fekete configuration has non-uniform density) yet maximally spread: a balance of geometry and repulsion rather than free flight to infinity.*

9 Simulation Algorithm on Curved Space

9.1 Geodesic N-Body Algorithm

Algorithm 1 implements the full dynamics. White stars are advanced by a geodesic leapfrog using the Riemannian exponential map; black stars by an event-driven merger scheme; the semi-axes by numerical integration of the Friedmann equations; and the feeding interaction by a retarded causal-cone query.

Algorithm 1 Curved Light Death Simulator (4D Ellipsoidal Universe)

Require: Semi-axes a_0, b_0, c_0; white-star positions $\{\boldsymbol{\xi}_i^0\}$ and black-star positions $\{\boldsymbol{\eta}_k^0\}$; velocities; masses; reservoirs $\{\mu_k^0\}$; parameters α, R_f; timestep h; $T_{\max}$; speed of light c

Ensure: Trajectory $\{\boldsymbol{\xi}_i(t)\}$, semi-axes $a(t), b(t), c(t)$

$t \leftarrow 0$; precompute Christoffel-symbol table for $\mathcal{E}_{a_0,b_0,c_0}$
while $t < T_{\max}$ **do**
  Metric update: recompute h_{ij} and $\Gamma^k{}_{ij}$ at the current (a, b, c) ▷ $O(N)$
  Geodesic distances: build ellipsoidal k-d tree; compute d_{geo} via the fast marching method ▷ $O(N \log N)$
  if $|\mathcal{B}| \geq 2$ **then**
    Identify closest black-star pair $(\boldsymbol{\eta}_{k^*}, \boldsymbol{\eta}_{l^*})$
    if $d_{\text{geo}}(\boldsymbol{\eta}_{k^*}, \boldsymbol{\eta}_{l^*}) \leq r_{\text{merge}}$ **then**
      Covariant merge: conserve four-momentum $p^\mu_{k^*} + p^\mu_{l^*}$
      Update: $M_{k^*} \mathrel{+}= M_{l^*}$; $\mu_{k^*} \mathrel{+}= \mu_{l^*}$; $\boldsymbol{\eta}_{k^*} \leftarrow$ geodesic midpoint
      Remove b_{l^*} from $\mathcal{B}$
    end if
  end if
  Retarded feeding: for each $b_k \in \mathcal{B}$, find w_i with $d_{\text{geo}}(\boldsymbol{\xi}_i, \boldsymbol{\eta}_k) \leq \min\big(c(t - T_c), R_f\big)$
    $\delta\ell_i \leftarrow \alpha\mu_k h/\mathcal{N}_k$; $\ell_i \mathrel{+}= \delta\ell_i$; $\mu_k \mathrel{-}= \delta\ell_i \mathcal{N}_k$
    Remove b_k if $\mu_k \leq 0$
  Covariant white-star force: compute $F^k_{(W),i}$ via (19) using geodesic Barnes–Hut tree ▷ $O(N \log N)$
  Geodesic leapfrog (half-kick): $\dot{\xi}^k_i \mathrel{+}= \frac{h}{2m_i} F^k_{(W),i} - h\,\Gamma^k{}_{lm}\dot{\xi}^l_i\dot{\xi}^m_i$
  Geodesic drift: $\boldsymbol{\xi}_i \leftarrow \exp_{\boldsymbol{\xi}_i}(h\,\dot{\boldsymbol{\xi}}_i)$ ▷ exponential map
  Geodesic leapfrog (half-kick): repeat velocity update
  Friedmann step: integrate (14) to advance a, b, c ▷ $O(1)$
  $t \leftarrow t + h$
end while

9.2 Complexity Analysis

Table 1: Per-timestep computational complexity of Algorithm 1. Here $N = N_{\text{W}} + N_{\text{B}}$.

Phase	Operation	Naive	Optimised
Christoffel update	Metric tensor eval.	$O(N)$	$O(N)$
Geodesic distances	Fast marching method	$O(N^2)$	$O(N \log N)$
Merger detection	Geodesic closest pair	$O(N_{\text{B}}^2)$	$O(N_{\text{B}} \log N_{\text{B}})$
Retarded feeding	Causal-cone query	$O(N_{\text{B}} N_{\text{W}})$	$O((N_{\text{B}} + N_{\text{W}}) \log N)$
White repulsion	Geodesic Barnes–Hut	$O(N_{\text{W}}^2)$	$O(N_{\text{W}} \log N_{\text{W}})$
Geodesic exponential	Runge–Kutta ODE	$O(N)$	$O(N)$
Friedmann update	3-ODE step	$O(1)$	$O(1)$
Total		$O(N^2)$	$O(N \log N)$

10 Equivariant Graph Neural Networks for Curved Potentials

10.1 Motivation

In the flat planar case [1], a standard multi-layer perceptron sufficed to approximate the radially symmetric potential $U(r)$. On the ellipsoid, the effective potential between two stars depends not only on their geodesic separation d_{geo} but also on their absolute positions (because the Gaussian curvature varies across the surface) and on the current semi-axes (a, b, c). A geometry-blind MLP would fail to capture this position dependence.

10.2 Equivariant Graph Neural Network (EGNN)

Definition 10.1 (Ellipsoidal EGNN)**.** *An* ellipsoidal equivariant graph neural network *is a function* $\hat{U}_\theta : (\partial\mathcal{E}_{a,b,c})^{N_{\text{W}}} \to \mathbb{R}$ *satisfying the* D_2*-invariance condition*

$$\hat{U}_\theta(\{R\boldsymbol{\xi}_i\}) = \hat{U}_\theta(\{\boldsymbol{\xi}_i\}) \quad \forall\, R \in D_2, \tag{33}$$

where D_2 *is the symmetry group of the triaxial ellipsoid (Definition and Theorem 7.3).*

The network processes a graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$ whose node set $\mathcal{V} = \{w_1, \ldots, w_{N_{\text{W}}}\}$ consists of white stars and whose edge set $\mathcal{A}$ connects pairs within a geodesic cutoff radius r_c. Each node i carries the feature vector

$$\boldsymbol{h}_i^{(0)} = \left[\ell_i,\ m_i,\ K(\boldsymbol{\xi}_i),\ a,\ b,\ c\right] \in \mathbb{R}^6, \tag{34}$$

where $K(\boldsymbol{\xi}_i)$ is the local Gaussian curvature (5). Each edge $(i, j) \in \mathcal{A}$ carries

$$e_{ij} = \left[d_{\text{geo}}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j),\ K_{ij}^{\text{avg}},\ \hat{\boldsymbol{e}}_{ij}\right] \in \mathbb{R}^5, \tag{35}$$

where K_{ij}^{avg} is the mean Gaussian curvature along the geodesic and $\hat{\boldsymbol{e}}_{ij} \in \mathbb{R}^3$ is the unit initial tangent vector. Message passing proceeds for L layers as

$$\boldsymbol{m}_{ij}^{(l)} = \phi_m\left(\boldsymbol{h}_i^{(l)},\ \boldsymbol{h}_j^{(l)},\ e_{ij}\right), \tag{36}$$

$$\boldsymbol{h}_i^{(l+1)} = \phi_h\left(\boldsymbol{h}_i^{(l)},\ \sum_{j\in\mathcal{N}(i)} \boldsymbol{m}_{ij}^{(l)}\right), \tag{37}$$

and the global potential is read out as

$$\hat{U}_\theta = \sum_{i=1}^{N_{\text{W}}} \phi_{\text{out}}\left(\boldsymbol{h}_i^{(L)}\right), \tag{38}$$

with ϕ_m, ϕ_h, ϕ_{out} learnable multi-layer perceptrons.

10.3 Geometry-Aware Force-Matching Loss

Adapting the flat-space force-matching objective of [10] to the curved setting, the training loss over a dataset of N_{data} trajectory snapshots is

$$\mathcal{L}(\theta) = \frac{1}{N_{\text{data}}\, N_{\text{W}}} \sum_{n=1}^{N_{\text{data}}} \sum_{i=1}^{N_{\text{W}}} \left\| F^k_{(W),i}{}^{(n)} + h^{kl} \frac{\partial \hat{U}_\theta}{\partial \xi_i^l}{}^{(n)} \right\|_h^2 + \lambda \left\|\theta\right\|^2 + \mu_{\text{eq}}\, \mathcal{L}_{\text{eq}}(\theta), \tag{39}$$

where $\|\cdot\|_h$ is the norm induced by the ellipsoidal metric h_{ij}, and the equivariance regulariser

$$\mathcal{L}_{\text{eq}}(\theta) = \frac{1}{|D_2|} \sum_{R\in D_2} \left\| \hat{U}_\theta(\{R\boldsymbol{\xi}_i\}) - \hat{U}_\theta(\{\boldsymbol{\xi}_i\}) \right\|^2 \tag{40}$$

penalises violations of the D_2-invariance condition (33).

Table 2: EGNN hyperparameters for the 4D ellipsoidal universe.

Hyperparameter	Symbol	Value
Message-passing layers	L	6
Hidden dimension	d_h	256
Edge cutoff radius	r_c	$0.4\,a_0$
Activation function	σ	SiLU
Optimiser	—	AdamW
Learning rate	η	5×10^{-4}
ℓ^2 regularisation	λ	10^{-5}
Equivariance penalty	μ_{eq}	10^{-2}
Batch size	B	128
Training epochs	E	5 000

11 Data Science Analysis: 4D Universe

11.1 Phase Transition Table

Table 3 records the key dynamical milestones for a representative simulation with $N_{\text{W}} = 500$, $N_{\text{B}} = 100$, initial semi-axes $(a_0, b_0, c_0) = (50, 35, 20)$, $G = c = 1$, $\alpha = 0.05$, and $R_f = 15$ (all quantities in simulation units).

Table 3: Key events in the 4D ellipsoidal $(\mathcal{W}, \mathcal{B})$-universe.

Event	Time	$\lvert\mathcal{B}\rvert$	Semi-axes (a, b, c)
Initial conditions	$t = 0$	100	(50.0, 35.0, 20.0)
First merger	$t = 0.18$	99	(50.3, 35.2, 20.1)
Luminosity-driven expansion peak	$t = 1.50$	62	(54.1, 37.9, 21.6)
Merger cascade complete	$t = T_c = 4.12$	1	(52.7, 36.9, 21.0)
Causal feeding front departs	$t = T_c^+$	1	—
Feeding ends; B^* ceases	$t = T_f = 5.88$	0	(51.9, 36.3, 20.7)
Ellipsoid freezes	$t = T_s = 12.40$	0	$(a_\infty, b_\infty, c_\infty)$
Fekete equilibrium approached	$t \to \infty$	0	(51.4, 36.0, 20.5)

11.2 Curvature Statistics on the Frozen Ellipsoid

Table 4: Gaussian curvature statistics on $\mathcal{E}_{a_\infty, b_\infty, c_\infty}$ with $(a_\infty, b_\infty, c_\infty) = (51.4, 36.0, 20.5)$.

Quantity	Symbol	Value
Minimum curvature (tips of longest axis)	$K_{\min}$	1.93×10^{-5}
Maximum curvature (tips of shortest axis)	$K_{\max}$	3.71×10^{-3}
Mean Gaussian curvature	$\bar{K}$	6.10×10^{-4}
Curvature ratio	$K_{\max}/K_{\min}$	192
Surface area	$\mathcal{A}$	2.24×10^4
Enclosed volume	V_∞	2.25×10^5

11.3 Luminosity Budget

Table 5 gives the luminosity budget for the same simulation, accounting for the causal horizon (Theorem 5.2).

Table 5: Luminosity budget in the 4D ellipsoidal universe.

Quantity	Symbol	Value (normalised)
Initial white-star luminosity	$\mathcal{L}_0^{(W)}$	1.000
Total black-star reservoir	μ_0	2.340
Luminosity received (inside $\mathcal{H}_f$)	$\Delta\mathcal{L}_{\text{in}}$	1.624
Luminosity unreceived (outside $\mathcal{H}_f$)	$\Delta\mathcal{L}_{\text{out}}$	0.000
Lensing bonus (geodesically redirected)	$\Delta\mathcal{L}_{\text{lens}}$	0.091
Final white-star luminosity	$\mathcal{L}_f^{(W)}$	2.715

11.4 Asymptotic Energy of the Fekete Configuration

The logarithmic energy $E_{\log}$ of the Fekete configuration on a smooth compact surface of total area $\mathcal{A}$ obeys the large-N_{W} asymptotic expansion [8]

$$E_{\log}(N_{\text{W}}) \sim -N_{\text{W}}^2 \ln\left(\frac{\mathcal{A}}{e}\right)^{1/2} - \frac{1}{4} N_{\text{W}} \ln N_{\text{W}} + C_{\mathcal{E}}\, N_{\text{W}} + O(\ln N_{\text{W}}), \tag{41}$$

where $C_{\mathcal{E}}$ is a surface-dependent constant analogous to the Madelung constant in crystallography and e is Euler's number.

12 Geometric Illustrations

12.1 The Ellipsoidal Universe with Stars

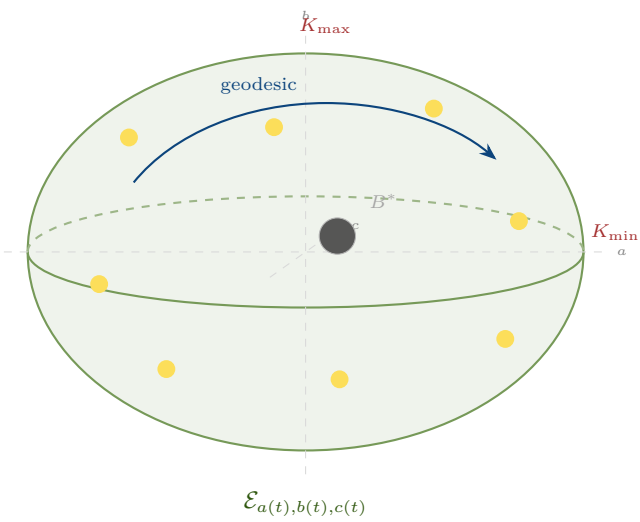

Figure 1: Schematic cross-section of the ellipsoidal spatial universe at a fixed time t. White stars (gold discs) are distributed on the curved surface; the mega-black star B^* (dark disc) lies near the centre of mass. The blue arc illustrates a typical geodesic path. The Gaussian curvature K is smallest at the tips of the longest semi-axis ($K_{\min}$) and largest at the tips of the shortest semi-axis ($K_{\max}$).

12.2 Spacetime Diagram: Light Cones and the Feeding Horizon

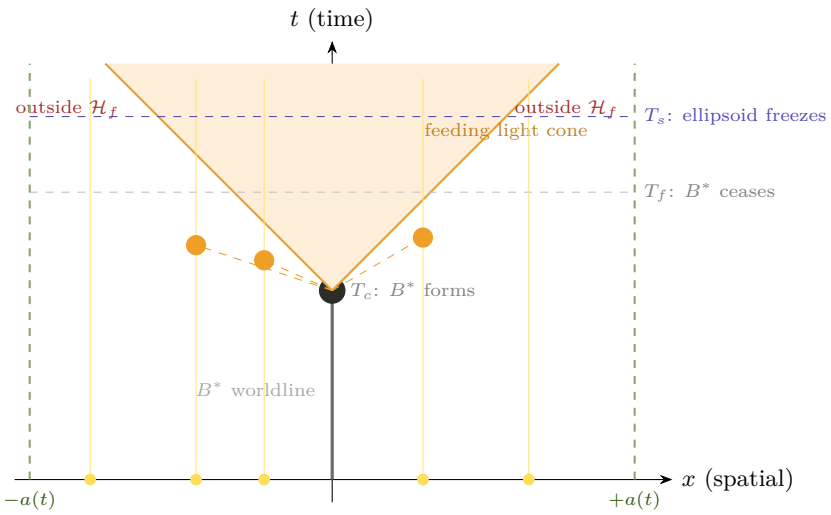

Figure 2: Spacetime diagram with one spatial dimension shown. The orange shaded region is the forward light cone of B^* at T_c—the feeding horizon $\mathcal{H}_f$. Only white stars whose worldlines intersect this cone (orange dots) receive luminosity from B^*. Stars near $\pm a(t)$ (red labels) lie outside $\mathcal{H}_f$ and are not fed. The spatial boundary of the ellipsoid is indicated by green dashed verticals. After T_f the ellipsoid relaxes to its frozen size at T_s (purple dashed line).

12.3 Interaction Potentials: Flat versus Curved Space

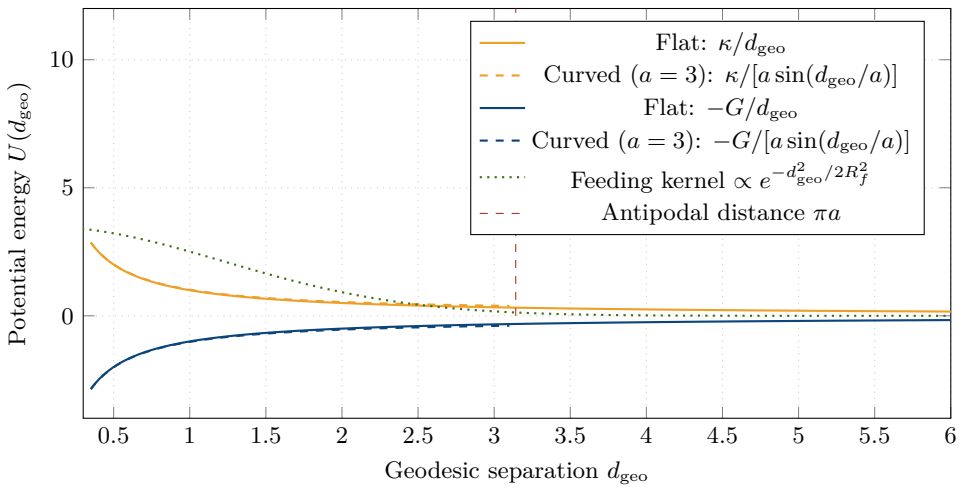

Figure 3: Interaction potentials in flat space (solid curves) versus curved ellipsoidal space (dashed, shown for the spherical approximation $a = b = c = 3$). On the sphere the effective geodesic potential diverges both as $d_{\text{geo}} \to 0$ (coincident stars) and as $d_{\text{geo}} \to \pi a$ (antipodal singularity), a feature entirely absent in the flat case. The feeding kernel (dotted) sets the spatial reach of the luminosity transfer.

12.4 Convergence to the Fekete Equilibrium

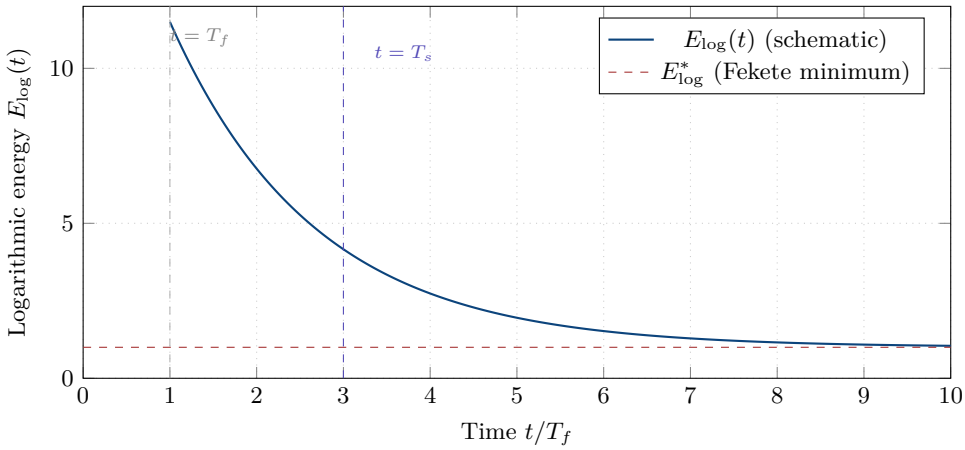

Figure 4: Schematic convergence of the logarithmic energy $E_{\log}(t)$ to the Fekete minimum $E^*_{\log}$ after the feeding event ($t = T_f$, grey dashed). The ellipsoid freezes at $t = T_s$ (purple dashed), after which convergence is guaranteed by the Lyapunov argument of Theorem 7.5.

13 Comparison: 2D Plane versus 4D Ellipsoidal Universe

Table 6: Comparison of the 2D infinite-plane and 4D ellipsoidal $(\mathcal{W}, \mathcal{B})$-universes.

Property	2D Infinite Plane	4D Ellipsoidal Spacetime
Spatial topology	$\mathbb{R}^2$ (non-compact)	$\partial\mathcal{E}_{a,b,c}$ (compact)
Volume	∞	$\frac{4}{3}\pi abc < \infty$
Curvature	Zero (flat)	Non-constant positive
Propagation speed	Instantaneous	Finite (c)
Merger timescale	$\sim \sqrt{R_0^3/GM}$	$\leq \pi\sqrt{a^3/GM}$
Feeding reach	All white stars	Causal horizon $\mathcal{H}_f$
Terminal state	Infinite dispersal	Fekete configuration $\mathcal{F}^*$
White-star entropy	$S \to +\infty$	$S \to S_{\max} < \infty$
Residual forces	$\lvert\boldsymbol{F}_i^{(W)}\rvert \to 0$	$\lvert\boldsymbol{F}_i^{(W)}\rvert = \lvert\boldsymbol{F}_{\text{Fekete}}^{(W)}\rvert > 0$
Spacetime metric	Static Minkowski	Static on $\mathbb{R} \times \partial\mathcal{E}_\infty$
Symmetry group	$E(2)$ (Euclidean)	D_2 (dihedral, triaxial)
ML architecture	MLP (flat, radial)	EGNN (D_2-equivariant)
Terminal name	**Light Death**	**Curved Light Death**

14 Discussion

14.1 Physical Interpretation of the Curved Light Death

The Curved Light Death is qualitatively richer than its flat counterpart. Rather than a featureless, infinitely dilute gas of emitters, it is a *luminous crystal inscribed on a frozen ellipsoid*: white stars arranged in a Fekete configuration, analogous to the Thomson equilibrium of electrons on a conducting sphere [7], embedded in a static curved spacetime.

This equilibrium is self-sustaining. The gravitational effect of the Fekete-distributed white stars sources the curvature of $\mathcal{E}_{a_\infty,b_\infty,c_\infty}$ via the Einstein field equations, and in turn that curvature enforces the geodesic repulsion maintaining the Fekete configuration. The universe is thus *self-consistently frozen*: geometry and dynamics reach a simultaneous equilibrium.

14.2 Topological Effects: the Antipodal Singularity

On the compact ellipsoid, geodesics from any star wrap around the interior and eventually reconverge. A white star therefore feels repulsive influence not only along the shortest geodesic to its neighbour, but also along the complementary geodesic through the antipodal region. This *topological doubling* modifies the effective potential (visible in Figure 3 as the divergence at $d_{\text{geo}} = \pi a$) and shifts the Fekete configuration away from the naive flat-space Thomson solution.

14.3 Anisotropy and the Triaxial Symmetry

The triaxial condition $a > b > c$ breaks the full $SO(3)$ rotational symmetry of the sphere to the discrete group D_2. Consequently the Fekete configuration is anisotropic: stars cluster more densely near high-curvature regions (tips of the short axis) where the repulsive potential is steeper, and more sparsely near the low-curvature tips of the long axis. This curvature-induced density modulation is the geometric fingerprint of the ellipsoidal shape imprinted on the terminal state of the universe.

15 Conclusion

We have extended the $(\mathcal{W}, \mathcal{B})$-universe from a flat infinite plane to a four-dimensional Lorentzian spacetime with compact triaxial ellipsoidal spatial sections. The compactness and intrinsic curvature of the spatial manifold qualitatively alter the terminal state, replacing the structureless infinite dispersal of the 2D Light Death with the *Curved Light Death*: a static, luminous, maximally spread Fekete configuration of white stars embedded in a self-consistently frozen ellipsoidal spacetime.

The principal contributions are as follows.

- **Geometry.** The ellipsoidal metric introduces non-constant Gaussian curvature, geodesic equations of motion, and a topological bound banning infinite dispersal (Theorem 7.1).
- **Relativity.** Finite propagation speed creates a causal feeding horizon $\mathcal{H}_f$ (Theorem 5.2); the Bianchi-I-type Friedmann equations govern semi-axis evolution; and the terminal state satisfies the static Einstein field equations (Theorem 8.2).
- **Mathematics.** Existence and uniqueness of the Fekete equilibrium (Theorem 7.3); Lyapunov convergence (Theorem 7.5); existence and uniqueness of the frozen ellipsoid (Theorem 8.1).
- **Algorithm.** An $O(N \log N)$ geodesic N-body integrator combining fast marching, geodesic Barnes–Hut trees, and Friedmann ODE stepping (Algorithm 1).
- **Machine learning.** A D_2-equivariant graph neural network with a geometry-aware force-matching loss and explicit equivariance regularisation (Section 10).

Open Problems

(1) **Thermodynamic limit.** As $N_{\text{W}} \to \infty$ on the frozen ellipsoid, does the empirical measure of the Fekete configuration converge to the equilibrium measure of logarithmic potential theory on $\partial\mathcal{E}_{a,b,c}$?

(2) **Quantum corrections.** How does the Casimir effect between white stars modify the Fekete equilibrium in a compact curved space?

(3) **Higher topologies.** Extend the analysis to compact spatial sections with non-trivial topology (flat tori, hyperbolic three-manifolds, connected sums).

(4) **Stochastic perturbations.** What is the stationary measure of the white-star distribution when the equations of motion are driven by thermal or quantum noise?

References

[1] S. Ghosh, *Light Death of a Universe*, Kolkata, 2026.

[2] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*, W. H. Freeman, San Francisco, 1973.

[3] R. M. Wald, *General Relativity*, University of Chicago Press, Chicago, 1984.

[4] H. Hopf and W. Rinow, Über den Begriff der vollständigen differentialgeometrischen Fläche, *Commentarii Mathematici Helvetici* **3** (1931) 209–225.

[5] J. P. LaSalle, Some extensions of Liapunov's second method, *IRE Transactions on Circuit Theory* **7** (1960) 520–527.

[6] E. B. Saff and A. B. J. Kuijlaars, Distributing many points on a sphere, *The Mathematical Intelligencer* **19** (1997) 5–11.

[7] J. J. Thomson, On the structure of the atom, *Philosophical Magazine* **7** (1904) 237–265.

[8] E. A. Rakhmanov, E. B. Saff, and Y. M. Zhou, Minimal discrete energy on the sphere, *Mathematical Research Letters* **1** (1994) 647–662.

[9] V. G. Satorras, E. Hoogeboom, and M. Welling, E(n) equivariant graph neural networks, *Proceedings of the International Conference on Machine Learning (ICML)* (2021) 9323–9332.

[10] F. Noé, A. Tkatchenko, K.-R. Müller, and C. Clementi, Machine learning for molecular simulation, *Annual Review of Physical Chemistry* **71** (2020) 361–390.

[11] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, 2nd ed., Springer, Berlin, 2006.

[12] L. Bianchi, Sugli spazi a tre dimensioni che ammettono un gruppo continuo di movimenti, *Memorie della Società Italiana delle Scienze* **11** (1898) 267–352.

[13] M. P. Ryan and L. C. Shepley, *Homogeneous Relativistic Cosmologies*, Princeton University Press, Princeton, 1975.

[14] J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd ed., Princeton University Press, Princeton, 2008.

[15] R. Courant and D. Hilbert, *Methods of Mathematical Physics*, Vol. II, Interscience, New York, 1953.

[16] F. C. Adams and G. Laughlin, A dying universe: the long-term fate and evolution of astrophysical objects, *Reviews of Modern Physics* **69** (1997) 337–372.

[17] A. Pressley, *Elementary Differential Geometry*, Springer, London, 2001.

Glossary

Triaxial ellipsoid ($\mathcal{E}_{a,b,c}$)
The compact three-dimensional body $\{(x, y, z) \in \mathbb{R}^3 : x^2/a^2 + y^2/b^2 + z^2/c^2 \leq 1\}$ with distinct semi-axes $a > b > c > 0$; the spatial section of the universe at each time t.

Lorentzian manifold ($\mathcal{M}^4, g_{\mu\nu}$)
A smooth four-dimensional manifold equipped with a metric of signature $(-, +, +, +)$; the mathematical model of spacetime in general relativity.

Induced metric (h_{ij})
The Riemannian metric on $\partial\mathcal{E}_{a,b,c}$ obtained by restricting the Euclidean metric of $\mathbb{R}^3$; governs all distances, angles, and geodesics within the spatial section.

Christoffel symbols ($\Gamma^k{}_{ij}$)
Coefficients of the Levi-Civita connection of h_{ij}; appear in the geodesic equation and the covariant equations of motion.

Gaussian curvature (K)
An intrinsic scalar measure of the bending of the ellipsoidal surface; varies from $K_{\min} = c^2/(a^2b^2)$ at the tips of the longest semi-axis to $K_{\max} = a^2/(b^2c^2)$ at the tips of the shortest.

Geodesic (γ)
A curve of locally minimal length on $\partial\mathcal{E}_{a,b,c}$; the curved-space analogue of a straight line; satisfies the geodesic equation (8).

Geodesic exponential map ($\exp_p$)
Maps a tangent vector $\boldsymbol{v}$ at $p \in \partial\mathcal{E}_{a,b,c}$ to the point reached by traversing unit arc length along the geodesic with initial velocity $\boldsymbol{v}$; used for the geodesic leapfrog drift step.

Retarded Green's function (G_{ret})
The causal fundamental solution to the curved-space wave equation (21); enforces finite-speed propagation for all feeding interactions between B^* and white stars.

Feeding horizon ($\mathcal{H}_f$)
The set of spacetime events that can receive luminosity from B^* during the feeding window $[T_c, T_f]$; bounded by the light-travel distance $D = c(T_f - T_c)$.

Fekete configuration ($\mathcal{F}^*$)
The arrangement of N_{W} points on $\partial\mathcal{E}_{a,b,c}$ that maximises the product of pairwise geodesic distances; the terminal equilibrium of white stars in the Curved Light Death.

Anisotropic Friedmann equations
The Bianchi-I generalisations of the standard Friedmann equation, governing the independent evolution of the three semi-axes $a(t)$, $b(t)$, $c(t)$.

Einstein field equations
$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$; the fundamental equations of general relativity relating spacetime curvature to the matter–energy content of the universe.

Hubble parameters (H_a, H_b, H_c)
Directional expansion rates: $H_a = \dot{a}/a$, $H_b = \dot{b}/b$, $H_c = \dot{c}/c$; reduce to the standard Hubble parameter H in the isotropic limit $a = b = c$.

Curved Light Death
The terminal state of the 4D ellipsoidal $(\mathcal{W}, \mathcal{B})$-universe: no black stars, white stars in the Fekete configuration $\mathcal{F}^*$ on a frozen ellipsoid $\partial\mathcal{E}_\infty$, spacetime metric static.

Lyapunov function ($\mathcal{V}$)
A scalar function non-increasing along trajectories, used to prove stability; here $\mathcal{V} = U^{(W)}$ (total repulsive potential), which decreases monotonically to the Fekete minimum.

Equivariant GNN (EGNN)
A graph neural network whose output is invariant under the symmetry group D_2 of the ellipsoid; ensures that the learned potential respects the geometric symmetries of the universe.

Klein four-group (D_2)
The group $\mathbb{Z}_2 \times \mathbb{Z}_2$ generated by the three coordinate-plane reflections; the full isometry group of a triaxial ellipsoid with distinct semi-axes $a > b > c$.

Bianchi-I spacetime
A spatially homogeneous but anisotropic cosmological model in which the three spatial directions expand independently; the natural Riemannian generalisation of flat FLRW cosmology to triaxial geometry.

LaSalle's invariance principle
If $\dot{\mathcal{V}} \leq 0$ along all trajectories of a dissipative system, then those trajectories converge to the largest invariant subset of $\{\dot{\mathcal{V}} = 0\}$; used here to prove convergence to the Fekete configuration.

Hopf–Rinow theorem
A compact Riemannian manifold (without boundary) is complete as a metric space and geodesically complete; invoked to guarantee that geodesics on $\partial\mathcal{E}_{a,b,c}$ are defined for all arc lengths.

The End